

Samuel Atkins

BSc, MPhys – The University of Leeds

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[Linked In](#)

Personal Profile

4th year BSc, MPhys student with a strong interest in experimental research. Experienced in designing and conducting laboratory experiments and analysing the often large datasets obtained using Python. Confident in presenting my findings through written reports and presentations.

Education

University of Leeds, Sept 2023 – present

Integrated Masters in Physics (BSc, MPhys)

Relevant Year 3 Modules	Result	Relevant Year 2 Modules	Result
Advanced Techniques in Experimental Physics	74	Experimental Physics	81
Nuclear Operations	70	Computing 2	65
Magnetism in Condensed Matter	64	Physics 4	55
Molecular Simulation: Theory and Practice	71	Physics 3	72
Quantum Matter	73	Maths 3	79
Machine Learning Approaches to Scientific Data Analysis	65	Stellar Structure and Evolution	73

Bedford Modern School 2015-2023

A-Levels: Mathematics – A, Physics – A, Chemistry – B

GCSE's: 10 GCSE's, grades 9-5 including Physics (9), Chemistry (9), Biology (9), Mathematics (8), English (7/7)

Academic and Personal Projects

Bragg Summer Internship – Measuring Work Function of Non-Metallic Materials Using KPFM

- Used KPFM techniques to map the surface potential and calculate the work function of thin polymer films
- Fabricated new films via spin coating and investigated the effects of film thickness and substrate material on work function measurements
- Produced a poster to present my findings and a detailed write up of techniques and results for future project students to use

Experimental Physics Projects

- Carried out several experiments in a variety of laboratories including Single Particle Tracking, Fast Field Cycling NMR and Low Temperature Electron Transport
- Analysed all data using my own python code
- Created presentations at the end of each project talking about each detail of the projects from data collection and analysis to further reading to enhance my knowledge on the topic

Science Outreach

- Travelled to a selection of local Junior/Lower schools to give presentations and demonstrations on various topics ranging from lasers and acid-base reactions to exothermic reactions and using microscopes.
- Presented to up to 50 students at a time asking interactive questions and walking them through their own experiments

Technical Skills and Online Courses

- AFM and KPFM
- Rapid Prototyping Laboratory
- Cryogens
- Fast field cycling NMR
- Python for analysis of various forms of experimental results
- Large range of data collection and analysis software

- Open Learn: Introduction to complex analysis (level 3)
- Open Learn: Introduction to computational thinking (level 2)
- Open Learn: Health and Safety in the Laboratory and Field (level 1)

Key Employability Skills

- **Time Management – As a Lifeguard:** Managing the time doing setups, cleaning and being on poolside alongside other tasks that needed finishing within a certain time frame
- **Communication – Whilst working as a Lifeguard and Leisure Assistant** it was crucial to communicate what setups needed to be or had been done, details of the pool users and serving the customers in an appropriate manner
- **Efficiency** – Finding the most efficient way to complete a task, for example, completing sports setups as a leisure assistant or taking measurements during experiments
- **Presenting** – Presented many experimental reports to a high level within my 3rd year Advanced Techniques In Experimental Physics module. Created stands and presentations on my research for school science fairs, delivered many sports reports whilst representing the school
- **Dedication** – Achieved Grade 8 Flute performance, rowing training up to 10 times per week as part of rowing team from 2018-2023 and winning a race a Henley Royal Regatta.

Work Experience

Jun-24 to Dec-24: Age Restriction Auditor, Serve Legal

- Carrying out up to eight age restriction tests in shops and supermarkets per day and preparing reports on my findings and the outcomes
- Requires attention to detail, time management and confidence

Apr-23 to Jul-23: Part of the Science Outreach Program for my school

- Planned and gave demonstrations and talks on scientific topics such as lasers to Lower/Junior School students
- Travelled to local schools as well as having school students come to my school for us to give these demonstrations

Mar-21 to Aug-23: Lifeguard and Leisure Assistant, Inspire All Leisure

- Ensured the safety of up to 48 pool users and all centre users, gave first aid where necessary, set up sports equipment and cleaned the centre
- Completed monthly Skills assessments and training on various scenarios that may occur within the centre and first aid

Achievements and Personal Interests

- Placed 21st in the experienced category and 7th in the novice category at the North of England University Archery League indoor championships for recurve bow type
- Sports full colours, rowing, three years in the school's 1st VIII (2021-2023) and competed in events such as Henley Royal Regatta, National Schools Regatta and Schools Head of the River
- ABRSM Grade 8 Flute Performance
- Avid nature photographer and was shortlisted for the RSPCA young photographers award in 2021 ([News article link](#))